

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

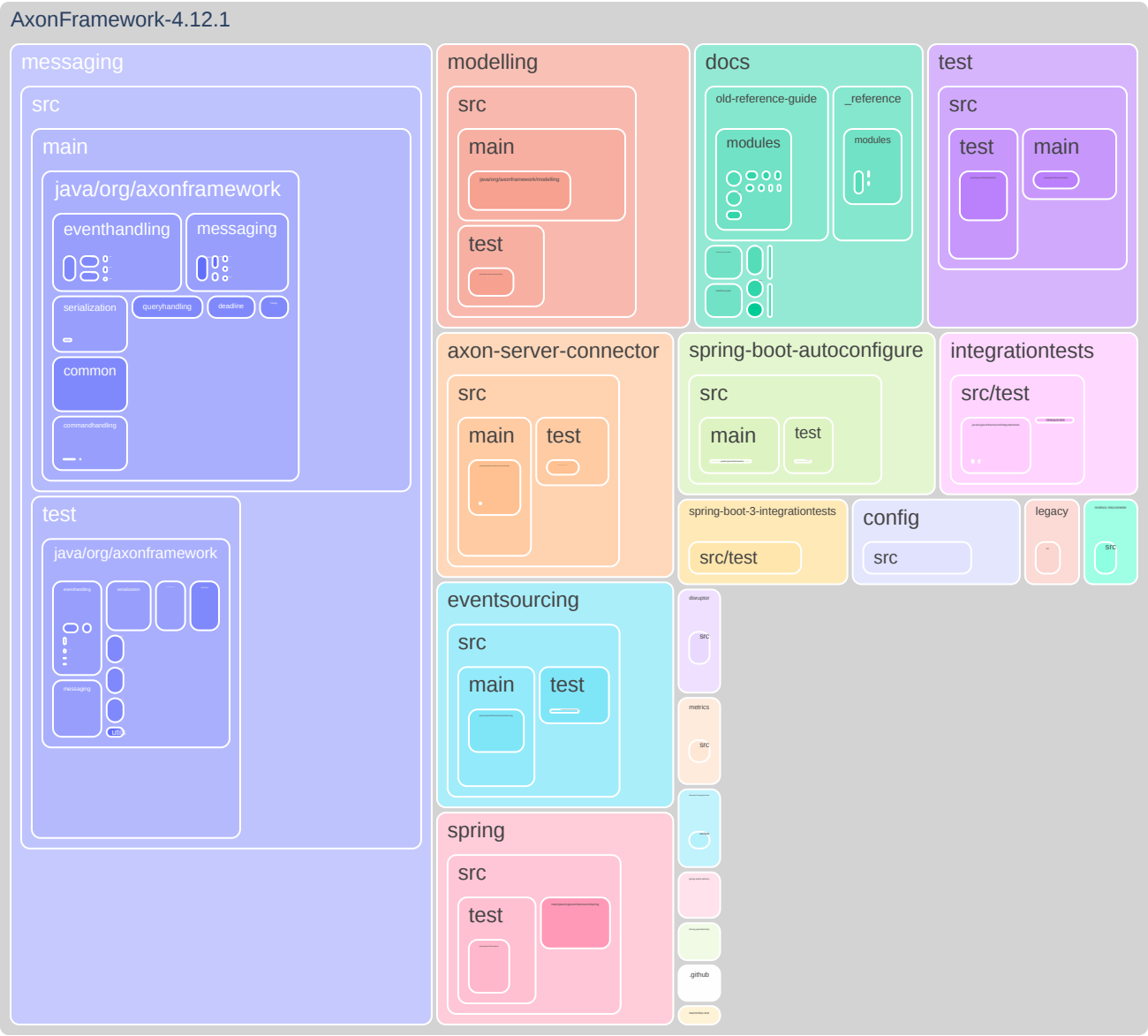
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	564.000000	564.000000	564.000000	564.000000	564.000000	564.000000
mean	24.898936	19.664894	288.570922	306.186170	1183.251773	400.652482
std	120.739280	21.297173	639.823202	170.624646	980.461529	392.566302
min	1.000000	2.000000	3.000000	28.000000	42.000000	28.000000
25%	1.000000	7.000000	27.000000	168.000000	442.000000	258.000000
50%	4.000000	12.000000	82.500000	401.000000	961.000000	401.000000
75%	10.000000	25.000000	270.250000	401.000000	2007.750000	401.000000
max	2294.000000	213.000000	9815.000000	2551.000000	5679.000000	3030.000000

	directoryPath	directoryParentPat
0	AxonFramework-4.12.1/assembly	AxonFramework-4.12.
1	AxonFramework-4.12.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat/source	AxonFramework-4.12.1/axon-serve
2	AxonFramework-4.12.1/axon-server-connector/src/main/resources/META-INF/services	AxonFramework-4.12.1/axon-server-connector/src/main/resource
3	AxonFramework-4.12.1/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event/util	AxonFramework-4.12.1/axon-serve
4	AxonFramework-4.12.1/config/src/test/resources/META-INF	AxonFramework-4.12.1/config/src/test/resource
5	AxonFramework-4.12.1/coverage-report	AxonFramework-4.12.
6	AxonFramework-4.12.1/docs/_playbook/localLinks	AxonFramework-4.12.1/docs/_playboo
7	AxonFramework-4.12.1/docs/_reference/modules/ROOT/attachments	AxonFramework-4.12.1/docs/_reference/modules/ROO
8	AxonFramework-4.12.1/docs/_reference/modules/ROOT/partials	AxonFramework-4.12.1/docs/_reference/modules/ROO
9	AxonFramework-4.12.1/docs/_reference/modules/axon-server-connector/pages	AxonFramework-4.12.1/docs/_reference/modules/axon-server-connect
10	AxonFramework-4.12.1/docs/_reference/modules/axon-server-connector/partials	AxonFramework-4.12.1/docs/_reference/modules/axon-server-connect
11	AxonFramework-4.12.1/docs/_reference/modules/configuration/pages	AxonFramework-4.12.1/docs/_reference/modules/configuratio
12	AxonFramework-4.12.1/docs/_reference/modules/configuration/partials	AxonFramework-4.12.1/docs/_reference/modules/configuratio
13	AxonFramework-4.12.1/docs/_reference/modules/disruptor/pages	AxonFramework-4.12.1/docs/_reference/modules/disruptor
14	AxonFramework-4.12.1/docs/_reference/modules/disruptor/partials	AxonFramework-4.12.1/docs/_reference/modules/disruptor
15	AxonFramework-4.12.1/docs/_reference/modules/eventsourcing/pages	AxonFramework-4.12.1/docs/_reference/modules/eventssourcin
16	AxonFramework-4.12.1/docs/_reference/modules/eventsourcing/partials	AxonFramework-4.12.1/docs/_reference/modules/eventssourcin
17	AxonFramework-4.12.1/docs/_reference/modules/legacy/pages	AxonFramework-4.12.1/docs/_reference/modules/legac
18	AxonFramework-4.12.1/docs/_reference/modules/legacy/partials	AxonFramework-4.12.1/docs/_reference/modules/legac
19	AxonFramework-4.12.1/docs/_reference/modules/metrics/pages	AxonFramework-4.12.1/docs/_reference/modules/metric
20	AxonFramework-4.12.1/docs/_reference/modules/metrics/partials	AxonFramework-4.12.1/docs/_reference/modules/metric
21	AxonFramework-4.12.1/docs/_reference/modules/micrometer/pages	AxonFramework-4.12.1/docs/_reference/modules/micromete
22	AxonFramework-4.12.1/docs/_reference/modules/micrometer/partials	AxonFramework-4.12.1/docs/_reference/modules/micromete
23	AxonFramework-4.12.1/docs/_reference/modules/modeling/partials	AxonFramework-4.12.1/docs/_reference/modules/modelin
24	AxonFramework-4.12.1/docs/_reference/modules/spring/pages	AxonFramework-4.12.1/docs/_reference/modules/sprin
25	AxonFramework-4.12.1/docs/_reference/modules/spring/partials	AxonFramework-4.12.1/docs/_reference/modules/sprin
26	AxonFramework-4.12.1/docs/_reference/modules/springboot-starter/pages	AxonFramework-4.12.1/docs/_reference/modules/springboot-starte
27	AxonFramework-4.12.1/docs/_reference/modules/springboot-starter/partials	AxonFramework-4.12.1/docs/_reference/modules/springboot-starte
28	AxonFramework-4.12.1/docs/_reference/modules/test/pages	AxonFramework-4.12.1/docs/_reference/modules/tes
29	AxonFramework-4.12.1/docs/_reference/modules/test/partials	AxonFramework-4.12.1/docs/_reference/modules/tes

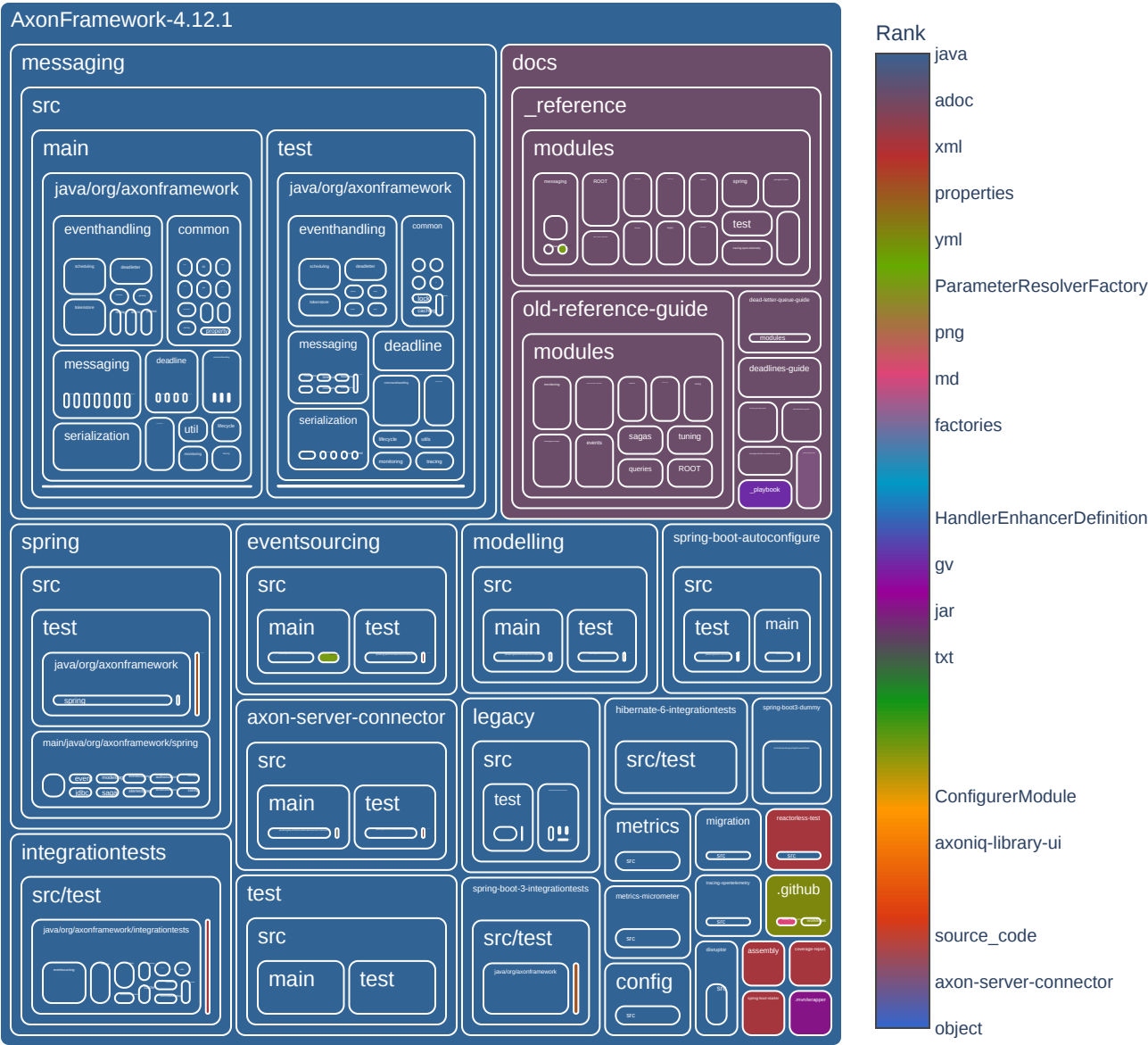
Number of files per directory

Directories and their file count



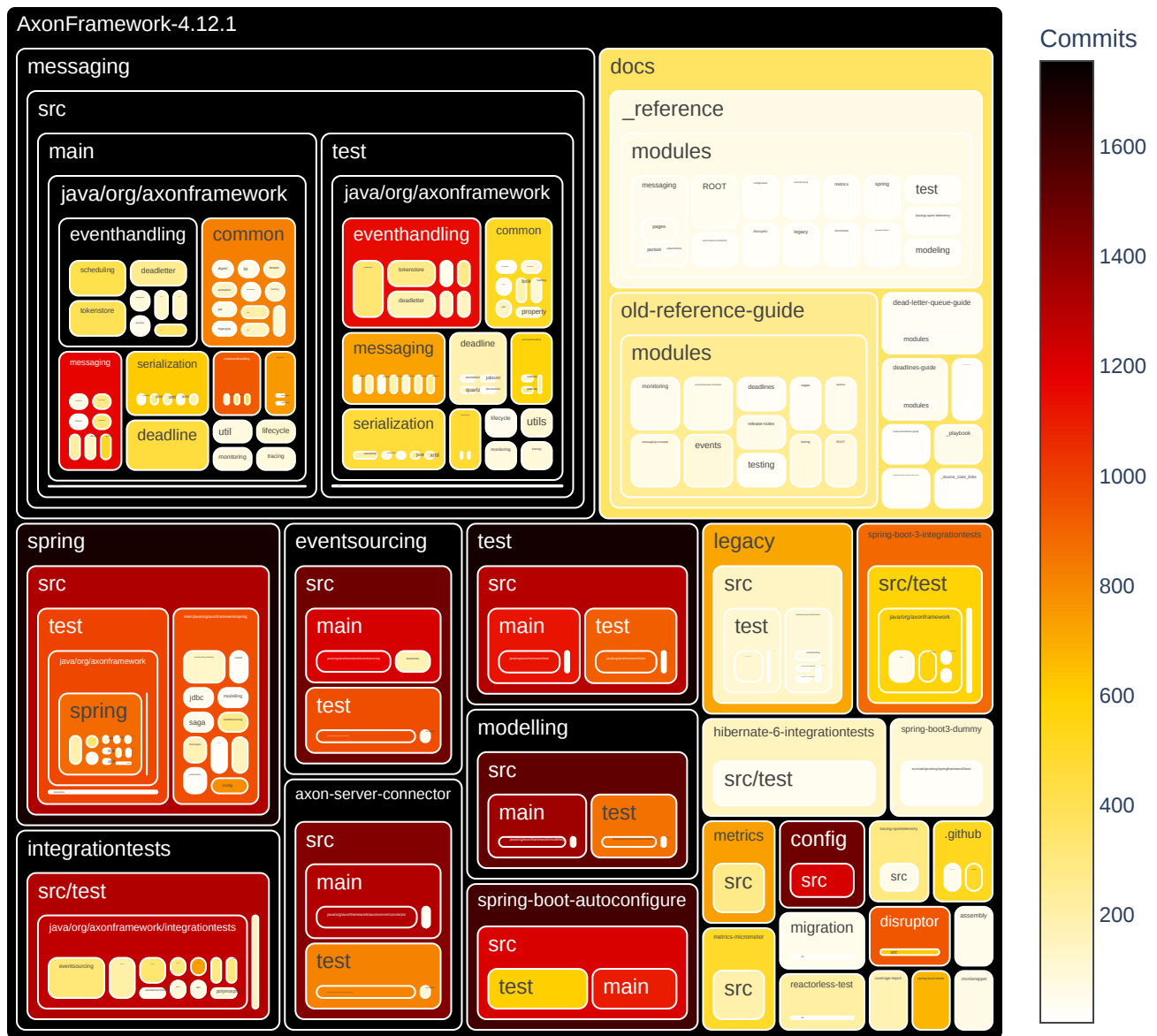
Most frequent file extension per directory

Most frequent file extension per directory



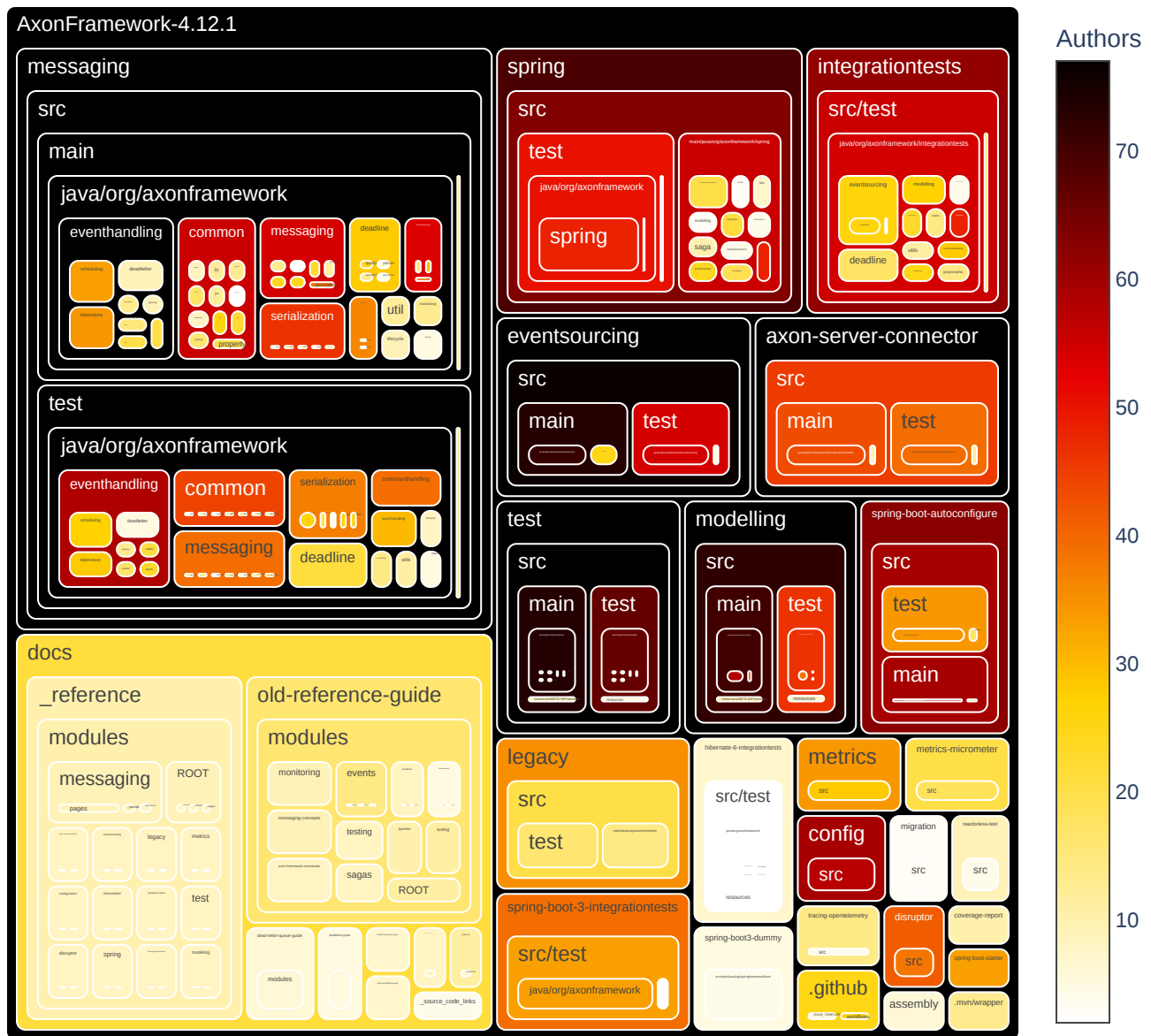
Number of commits per directory

Number of git commits



Number of distinct authors per directory

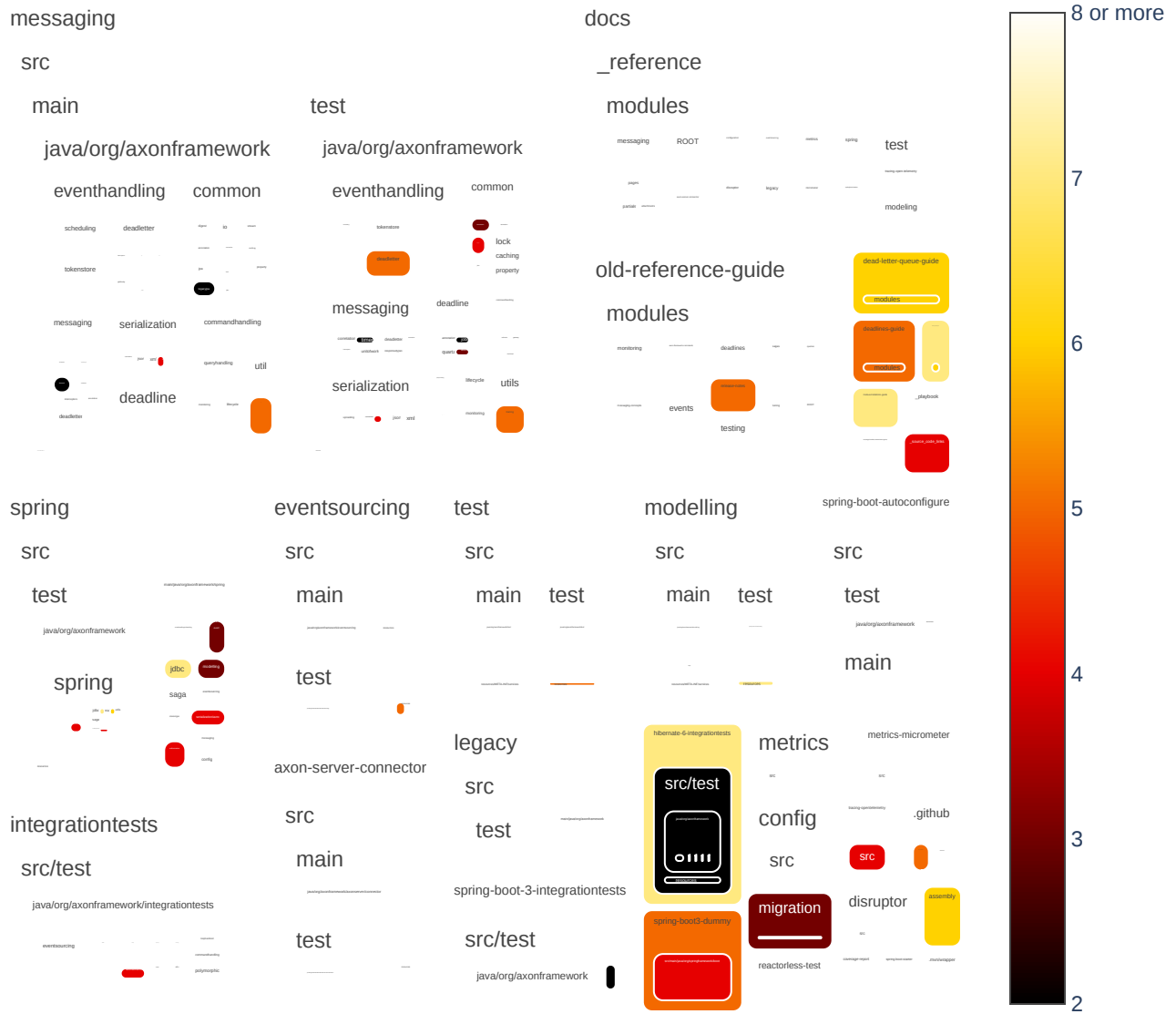
Number of distinct commit authors



Directories with very few different authors

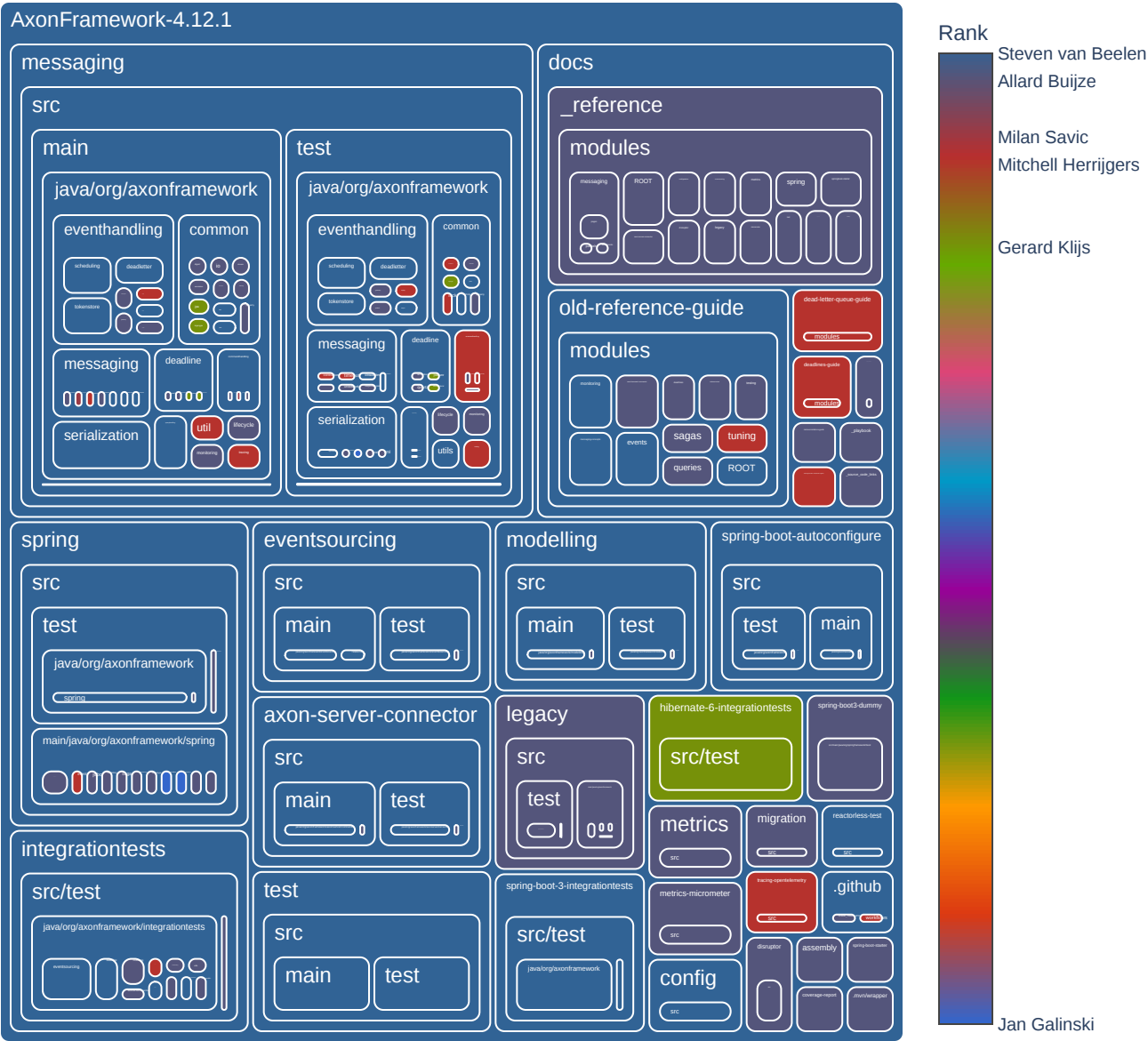
Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-4.12.1



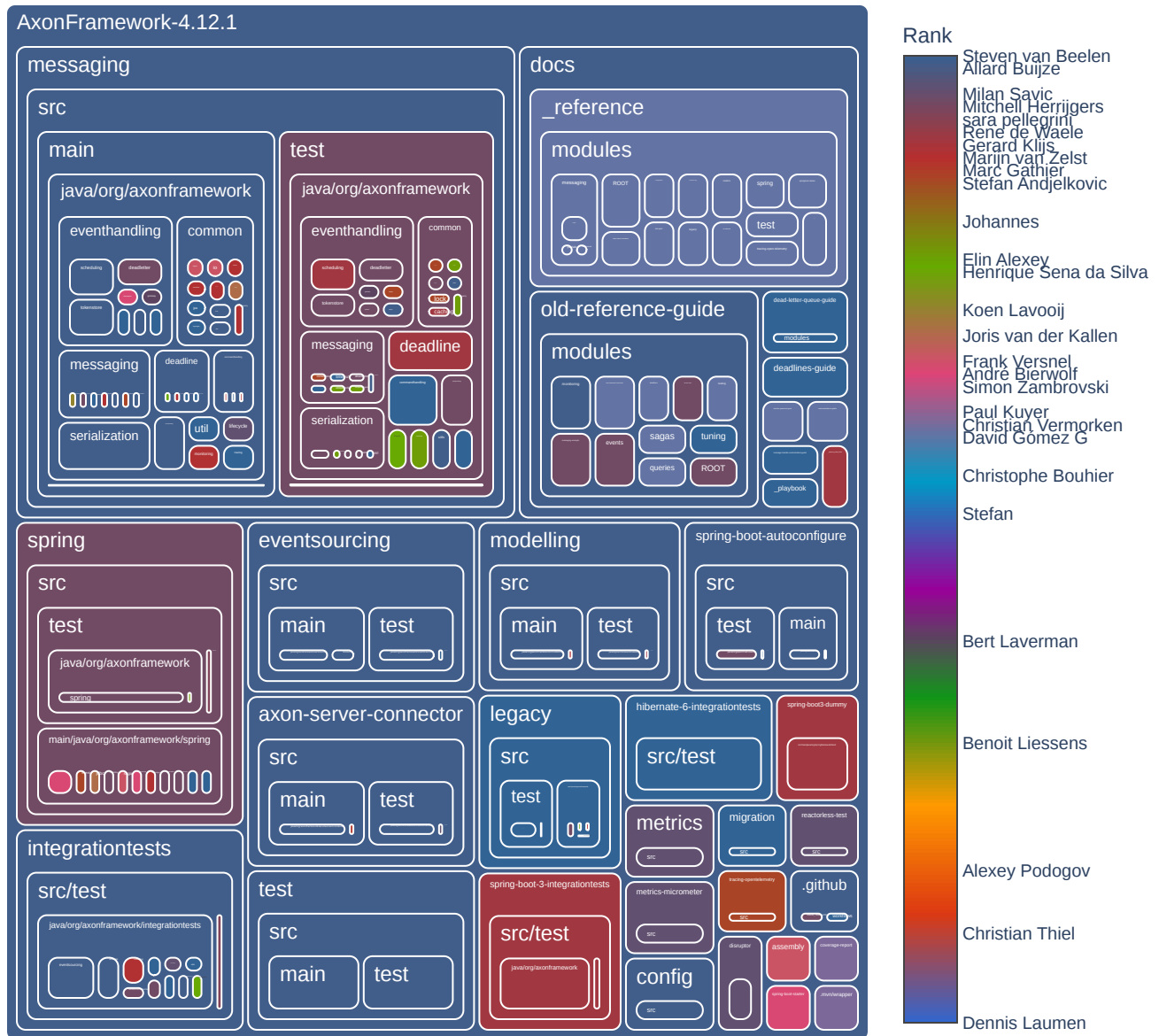
Main author per directory

Main authors with highest number of commits



Second author per directory

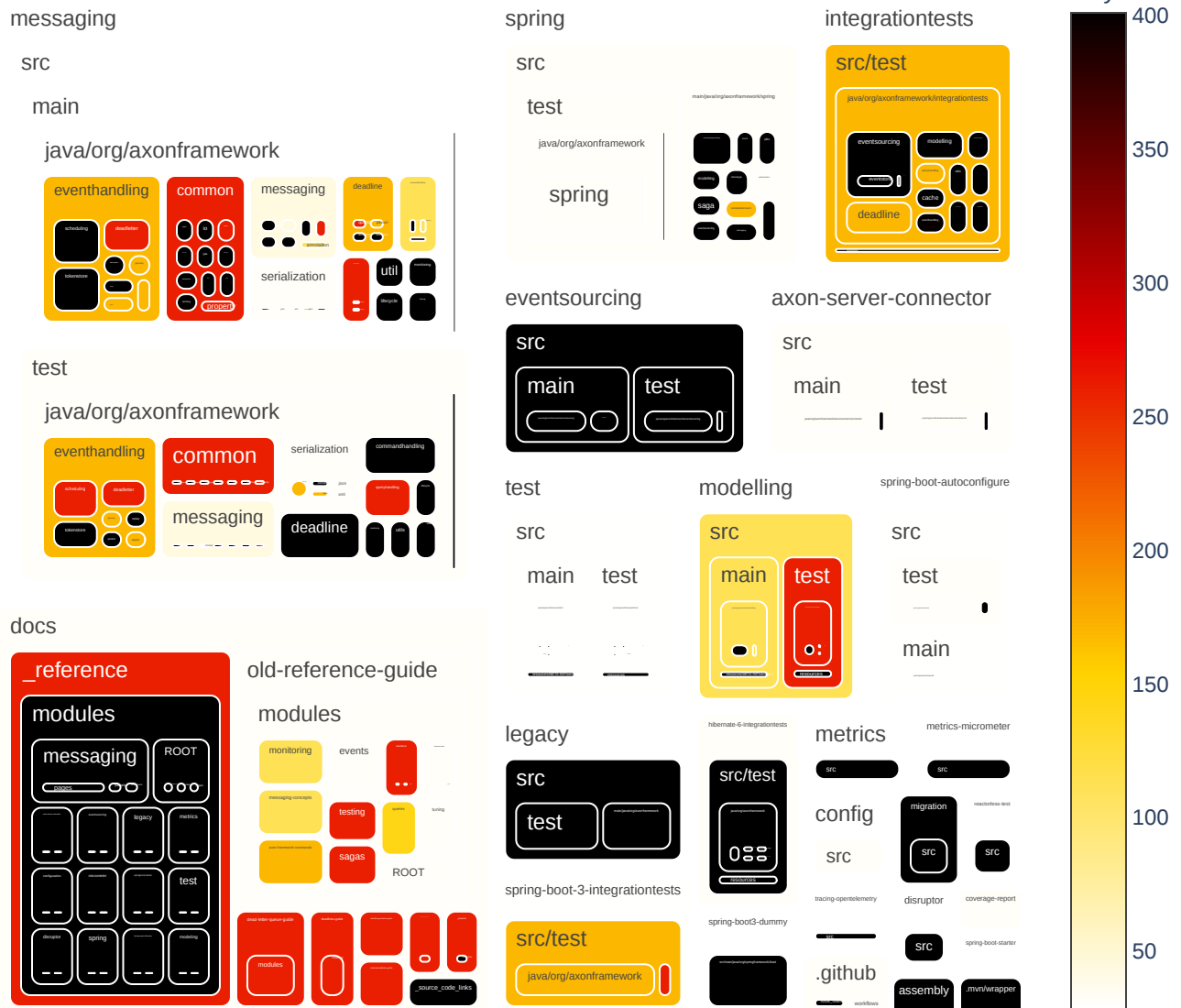
Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

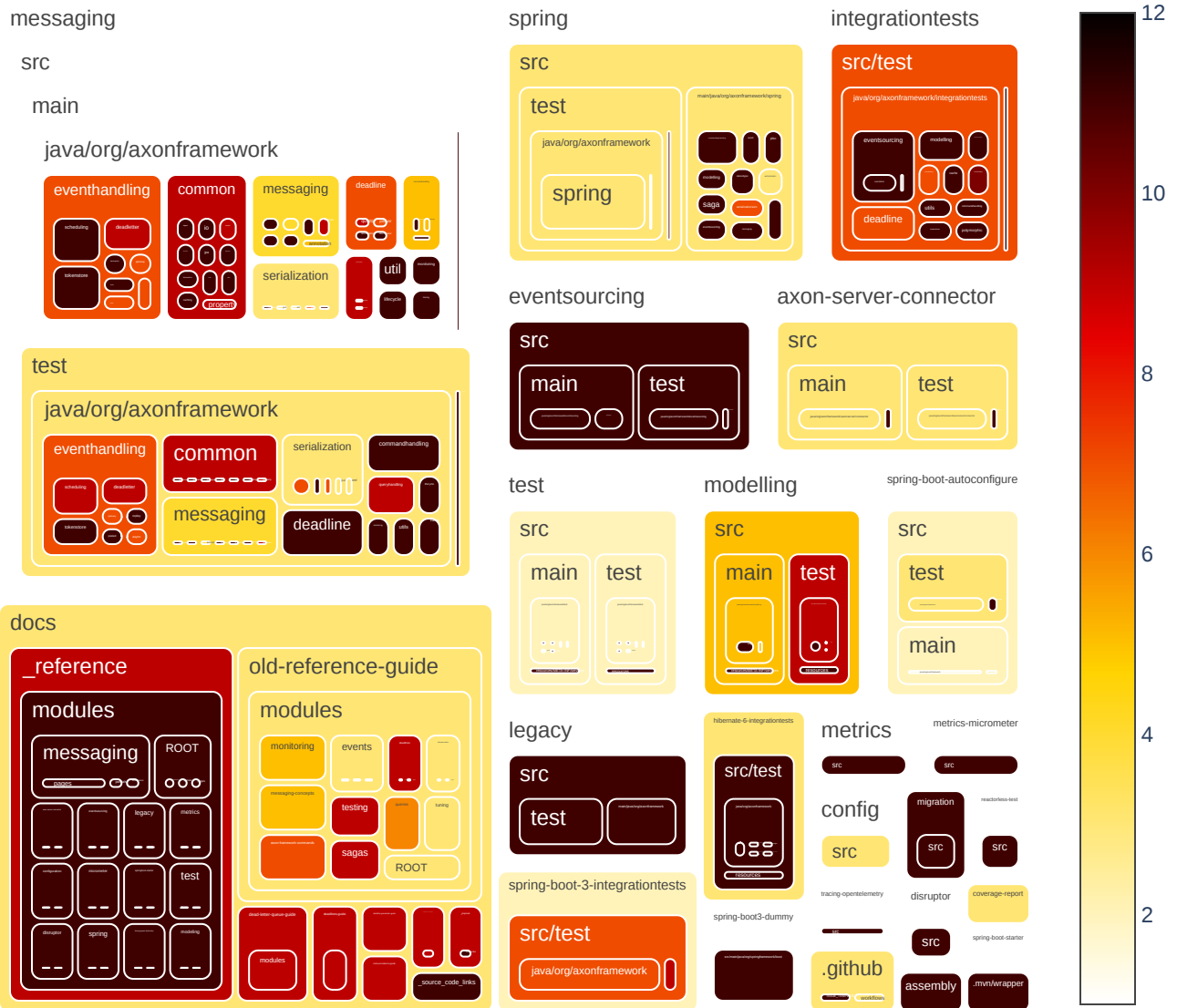
AxonFramework-4.12.1



Days since last commit per directory (ranked)

Rank of days since last commit

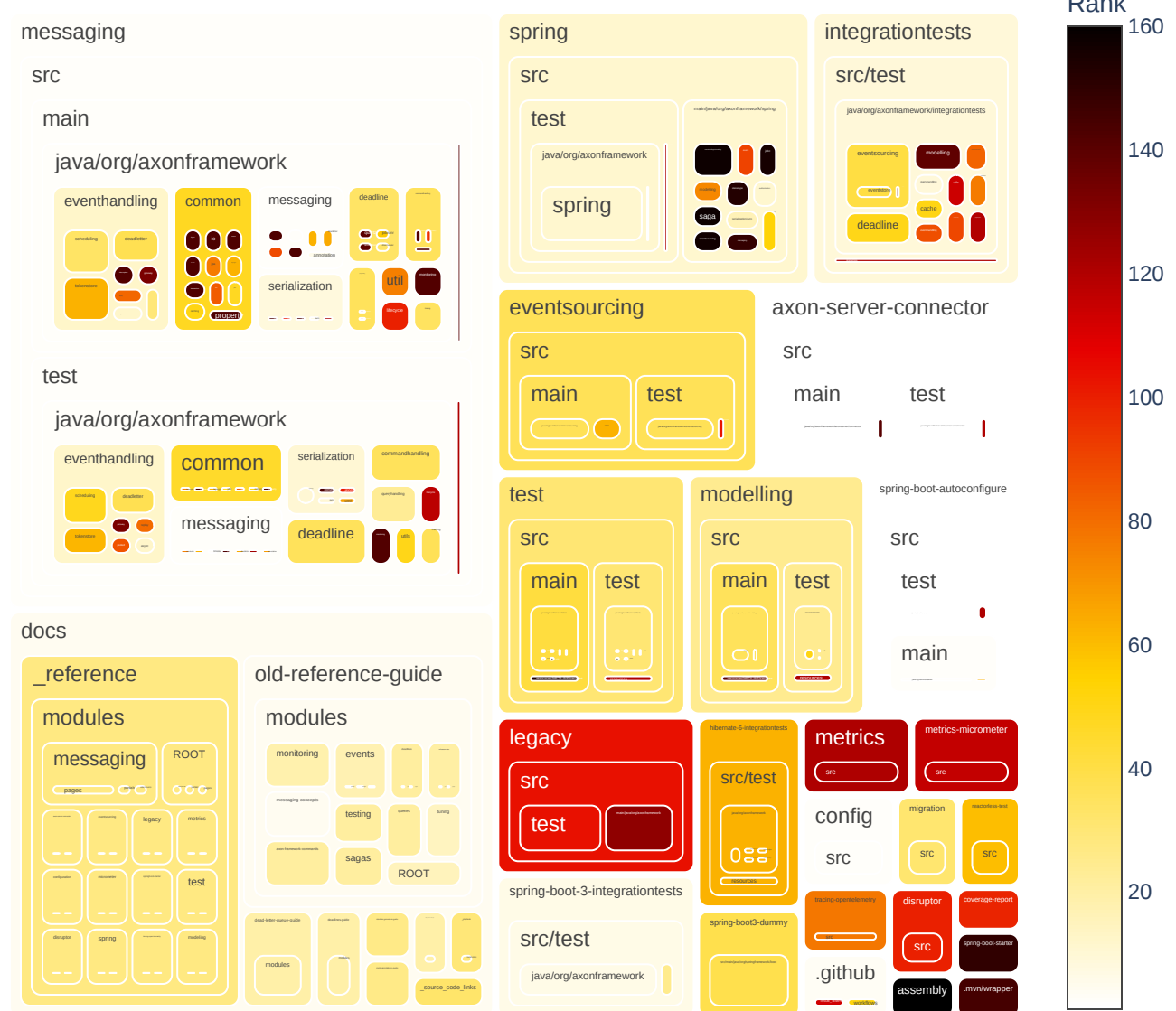
AxonFramework-4.12.1



Days since last file creation per directory

Rank of days since last file creation

AxonFramework-4.12.1



Days since last file modification per directory

Days since last file modification

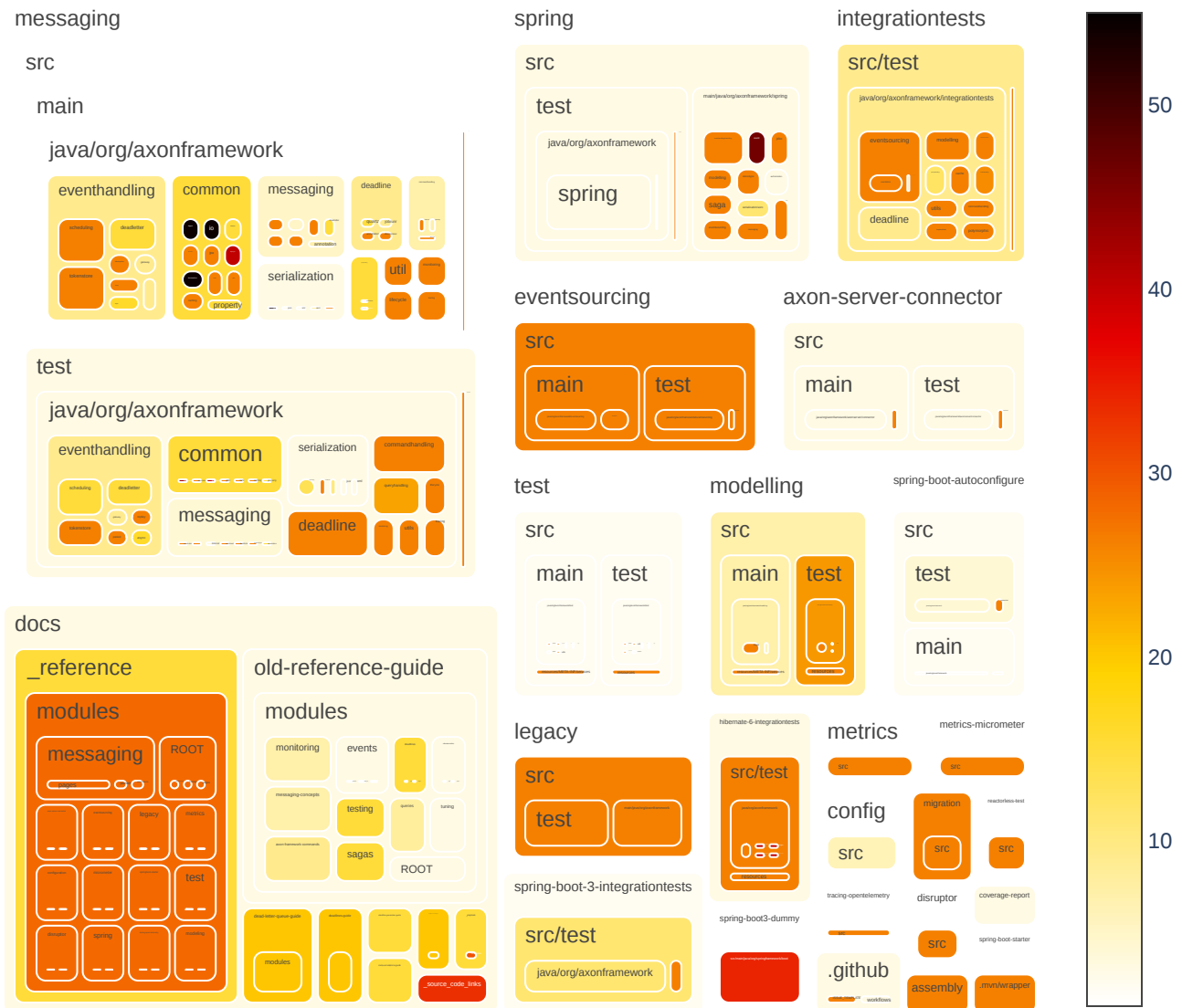
AxonFramework-4.12.1



Days since last file modification per directory (ranked)

Rank of days since last file modification

AxonFramework-4.12.1



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

Sum of commits that changed more than 30 files (each) = 973

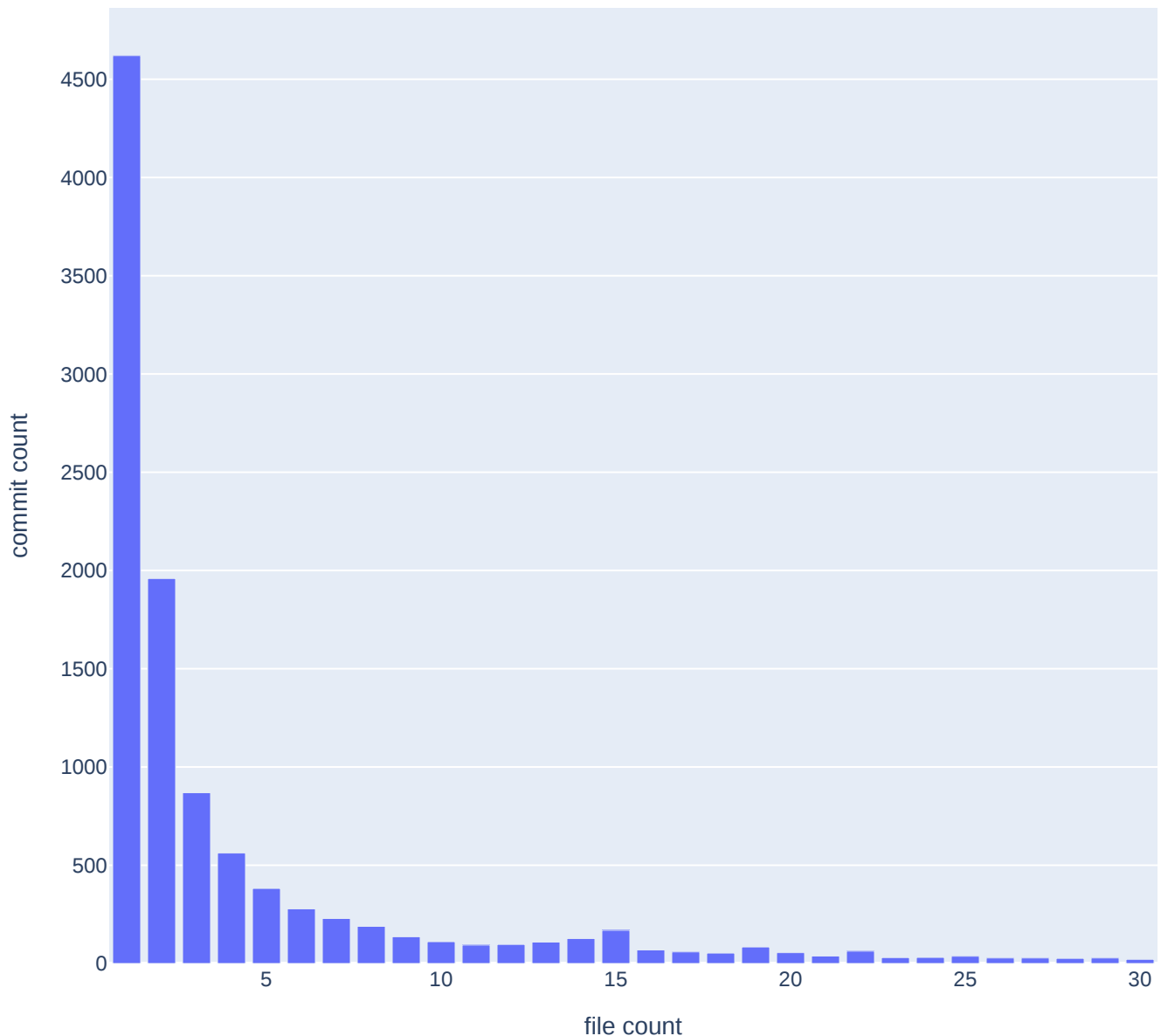
```
Max changed files with one commit = 2178
```

	filesPerCommit	commitCount
count	334.000000	334.000000
mean	343.449102	34.592814
std	439.862566	281.096056
min	1.000000	1.000000
25%	84.250000	1.000000
50%	185.000000	2.000000
75%	426.500000	5.000000
max	2178.000000	4620.000000

	filesPerCommit	commitCount
0	1	4620
1	2	1959
2	3	869
3	4	562
4	5	382
5	6	278
6	7	229
7	8	189
8	9	136
9	10	111
10	11	94
11	12	97
12	13	109
13	14	127
14	15	169
15	16	69
16	17	59
17	18	53
18	19	84
19	20	55
20	21	38
21	22	63
22	23	30
23	24	31
24	25	37
25	26	28
26	27	28
27	28	26
28	29	28
29	30	21

Bar chart with the number of files per commit distribution

Changed files per commit



Pairwise Changed Files

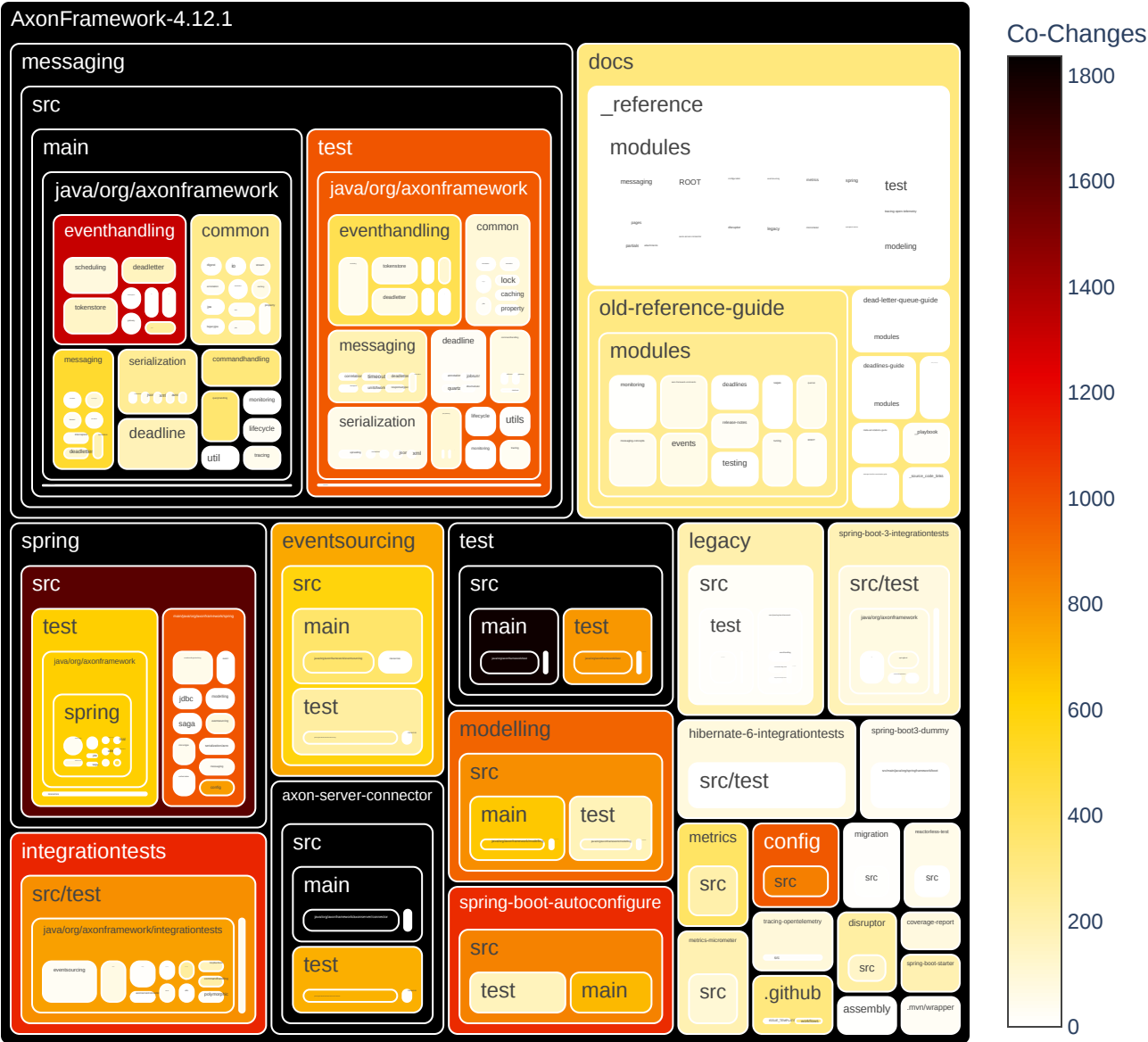
This section analyzes files that were changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

- **Commit Count:** The number of commits in which two files were changed together.
- **Commit Lift:** A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity:** The ratio of commits involving either file that also involved both files.

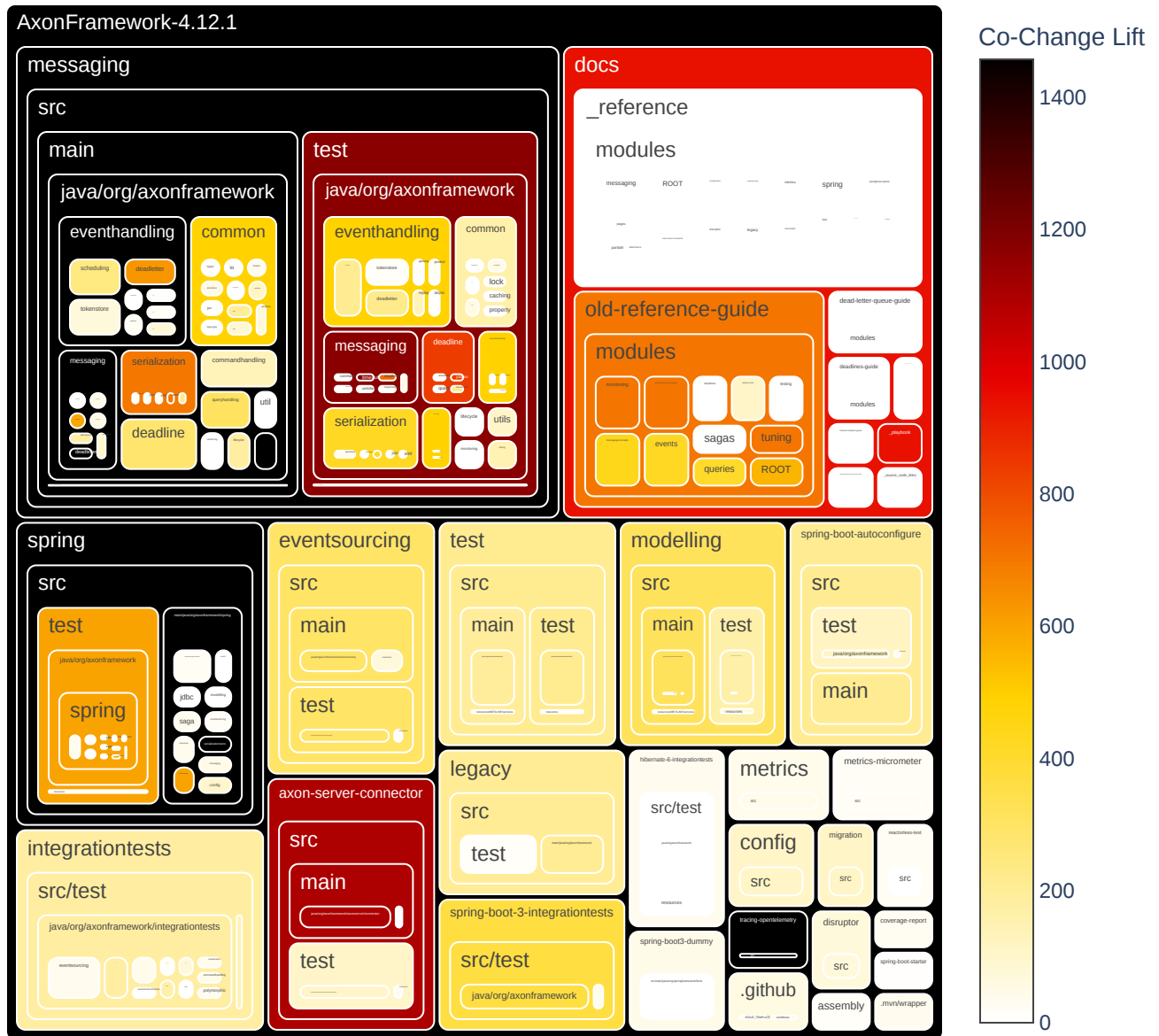
The following tables show the top pairwise changed files based on these metrics. The following charts show how these metrics are distributed across pairs of files that were changed together.

Treemap with files changed frequently with others

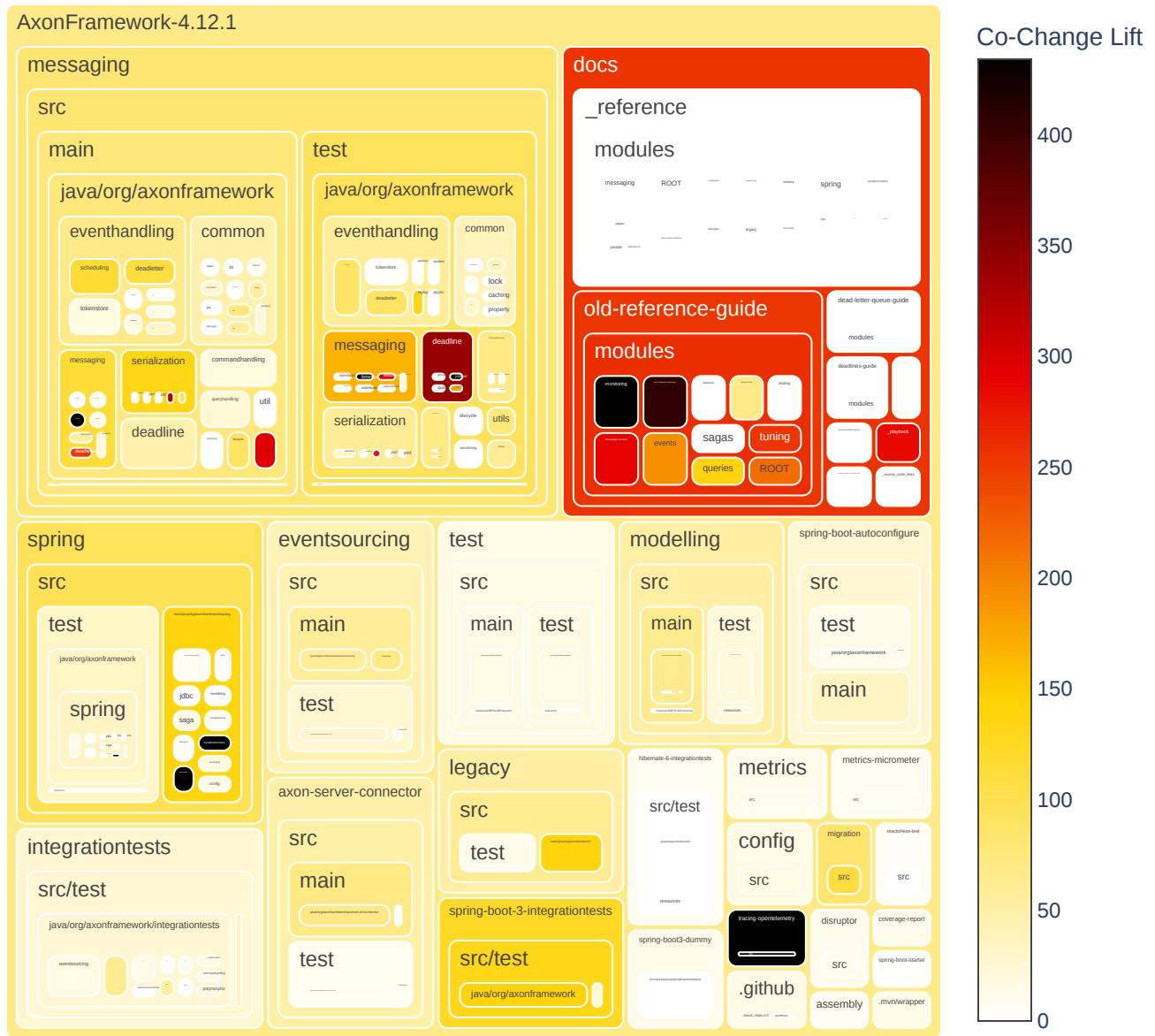
Files that likely co-change with others in update commits



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,

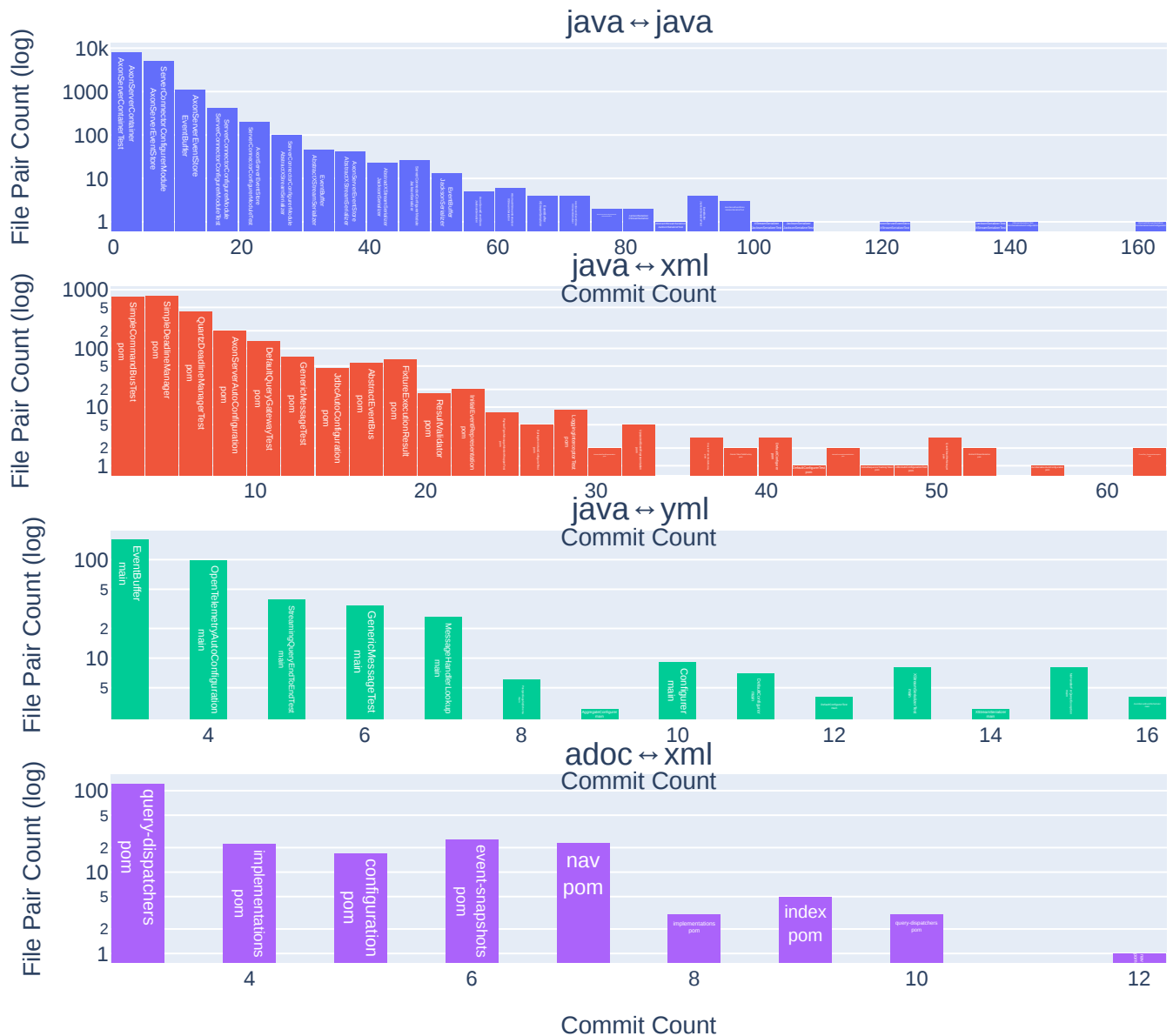


	fileExtensionPair	fileExtensionPairCount
0	java↔java	15189
1	java↔xml	2637
2	java↔yml	408
3	adoc↔xml	221

Files changed together by commit count

	fileExtensionPair	updateCommitCount	GroupRank	filePair
0	java↔java	163	1	AxonServerQueryBus↔AxonServerCommandBus
1	java↔java	144	2	AxonServerQueryBus↔AxonServerQueryBusTest
2	java↔java	135	3	SpringAxonAutoConfigurerTest↔SpringAxonAutoConfigurer spring/src/tes
3	java↔java	121	4	TrackingEventProcessorTest↔TrackingEventProcessor integrationtests/src/test/java/org/axonfra
4	java↔java	107	5	SpringAxonAutoConfigurer↔AxonConfiguration spi
5	java↔java	100	6	EventProcessingModule↔EventProcessingModuleTest
6	java↔java	96	7	SagaTestFixture↔AggregateTestFixture
7	java↔java	95	8	AxonServerQueryBusTest↔AxonServerCommandBusTest
8	java↔java	94	9	AggregateTestFixture↔FixtureConfiguration
9	java↔java	92	10	AggregateTestFixture↔SpringAxonAutoConfigurer test/
10	java↔xml	62	1	AggregateTestFixture↔pom
11	java↔xml	57	2	AggregateTestFixture↔pom
12	java↔xml	53	3	AggregateTestFixture↔pom
13	java↔xml	51	4	SpringAxonAutoConfigurer↔pom
14	java↔xml	48	5	SagaTestFixture↔pom
15	java↔xml	46	6	SpringAxonAutoConfigurer↔pom
16	java↔xml	45	7	SpringAxonAutoConfigurerTest↔pom
17	java↔xml	44	8	SpringAxonAutoConfigurerTest↔pom
18	java↔xml	42	9	AggregateTestFixture↔pom
19	java↔xml	41	10	AxonConfiguration↔pom
20	java↔yaml	16	1	JobRunrDeadlineManagerTest↔main
21	java↔yaml	15	2	JobRunrEventSchedulerTest↔main
22	java↔yaml	14	3	TrackingEventProcessor↔main
23	java↔yaml	13	4	SimpleQueryBus↔main
24	java↔yaml	12	5	SimpleEventHandlerInvokerTest↔main
25	java↔yaml	11	6	AxonServerEventStore↔main
26	java↔yaml	10	7	PooledStreamingEventProcessor↔main
27	java↔yaml	9	8	AxonServerQueryBusTest↔main
28	java↔yaml	8	9	DefaultConfigurer↔main
29	java↔yaml	7	10	AggregateAnnotationCommandHandler↔main
30	adoc↔xml	12	1	minor-releases↔pom
31	adoc↔xml	10	2	rdbms-tuning↔pom
32	adoc↔xml	9	3	minor-releases↔pom
33	adoc↔xml	8	4	minor-releases↔pom
34	adoc↔xml	7	5	minor-releases↔pom
35	adoc↔xml	6	6	nav↔pom
36	adoc↔xml	5	7	event-versioning↔pom
37	adoc↔xml	4	8	rdbms-tuning↔pom
38	adoc↔xml	3	9	query-dispatchers↔pom

Co-Changed Files by their commit count

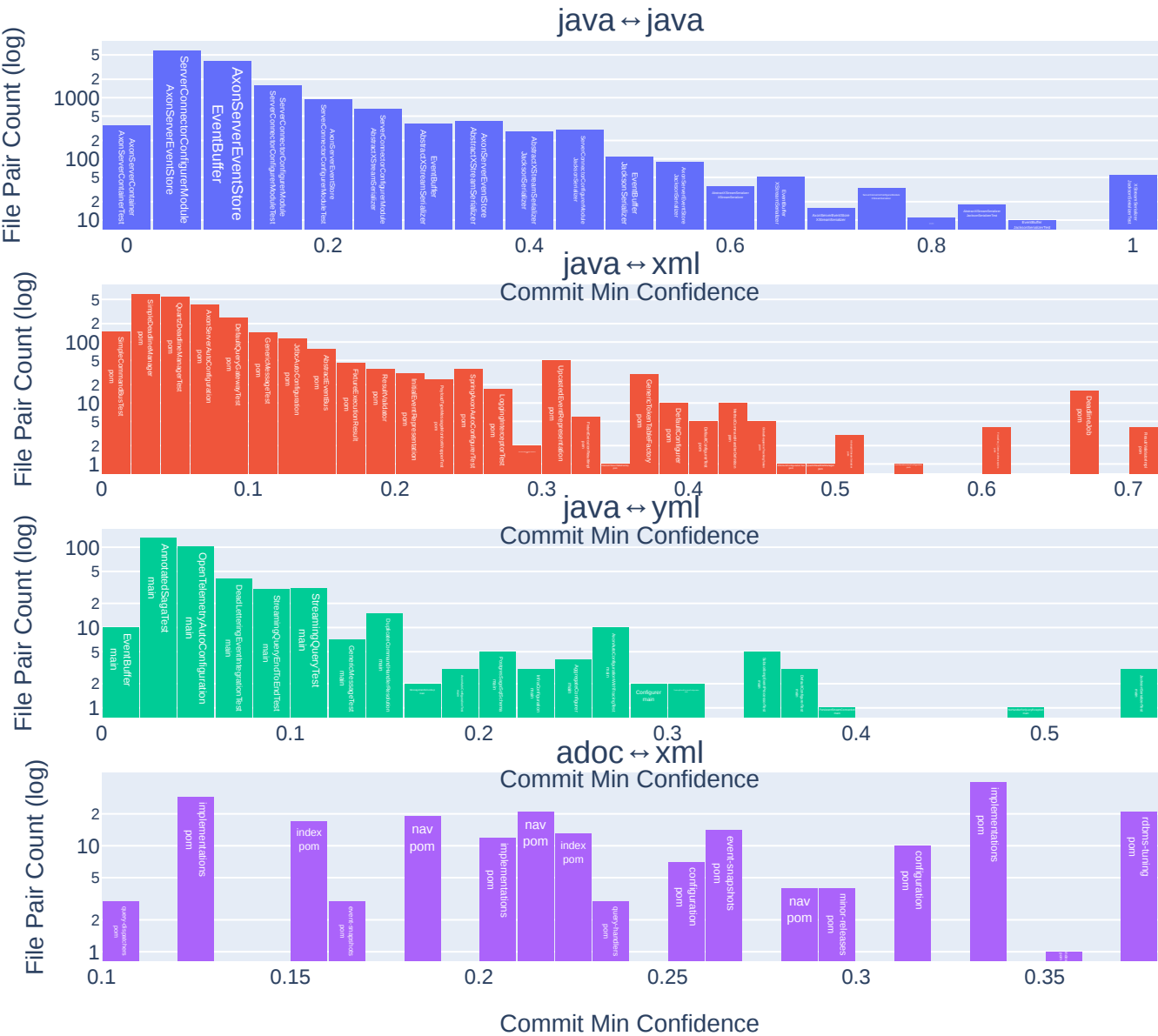


Files changed together by commit min confidence

The commit min confidence is the commit count where both files were changed divided by the commit count of the file with the least commits. This metric is useful to identify pairs of files that are frequently changed together and is not biased by single files that are changed very often.

	fileExtensionPair	updateCommitMinConfidence	GroupRank	filePair
0	java↔java	1.000000	1	AxonTimeLimitedTask↔UnitOfWorkTimeoutInterceptor
1	java↔java	0.909091	2	AvroUtil↔SpecificRecordBaseSerializerStrategy
2	java↔java	0.900000	3	AxonTimeLimitedTask↔TimeoutWrappedMessageHandlingMember
3	java↔java	0.875000	4	Decisions↔DoNotEnqueue
4	java↔java	0.857143	5	AvroSerializer↔AvroSerializerStrategy
5	java↔java	0.833333	6	AvroSerializerStrategy↔ByteArrayToGenericRecordConverter
6	java↔java	0.800000	7	DeadLetteringEventIntegrationTest↔InMemoryDeadLetteringIntegrationTest
7	java↔java	0.785714	8	AvroSerializer↔AvroSerializerTest
8	java↔java	0.777778	9	JobrunrDeadlineManagerTest↔JobRunrEventSchedulerTest
9	java↔java	0.750000	10	AxonServerAutoConfiguration↔PersistentStreamMessageSource
10	java↔xml	0.703704	1	JobRunrEventSchedulerTest↔pom
11	java↔xml	0.666667	2	JobRunrEventSchedulerTest↔pom
12	java↔xml	0.600000	3	ChainingConverter↔pom
13	java↔xml	0.545455	4	EventStoreTestUtils↔pom
14	java↔xml	0.512821	5	DefaultEventMessageConverter↔pom
15	java↔xml	0.500000	6	ChainingConverter↔pom
16	java↔xml	0.487179	7	DefaultEventMessageConverter↔pom
17	java↔xml	0.461538	8	DefaultEventMessageConverter↔pom
18	java↔xml	0.454545	9	JpaEventStoreAutoConfigurationWithoutAxonServerTest↔pom
19	java↔xml	0.448276	10	JobrunrDeadlineManagerTest↔pom
20	java↔yaml	0.555556	1	JobRunrEventSchedulerTest↔main
21	java↔yaml	0.481481	2	JobRunrEventSchedulerTest↔antora
22	java↔yaml	0.392857	3	AvroSerializerAutoConfiguration↔antora
23	java↔yaml	0.375000	4	AxonTimeLimitedTaskTest↔main
24	java↔yaml	0.357143	5	AvroSerializerAutoConfiguration↔main
25	java↔yaml	0.340909	6	AbstractDeadlineManagerTestSuite↔main
26	java↔yaml	0.300000	7	ChainingConverter↔main
27	java↔yaml	0.295455	8	AbstractDeadlineManagerTestSuite↔antora
28	java↔yaml	0.288889	9	JobrunrDeadlineManagerTest↔antora
29	java↔yaml	0.277778	10	SpringAxonConfiguration↔main
30	adoc↔xml	0.375000	1	index↔pom
31	adoc↔xml	0.352941	2	minor-releases↔pom
32	adoc↔xml	0.333333	3	nav↔pom
33	adoc↔xml	0.315789	4	query-dispatchers↔pom
34	adoc↔xml	0.312500	5	query-handlers↔pom
35	adoc↔xml	0.291667	6	event-snapshots↔pom
36	adoc↔xml	0.285714	7	implementations↔pom
37	adoc↔xml	0.281250	8	rbms-tuning↔pom
38	adoc↔xml	0.264706	9	minor-releases↔pom
39	adoc↔xml	0.263158	10	event-versioning↔pom

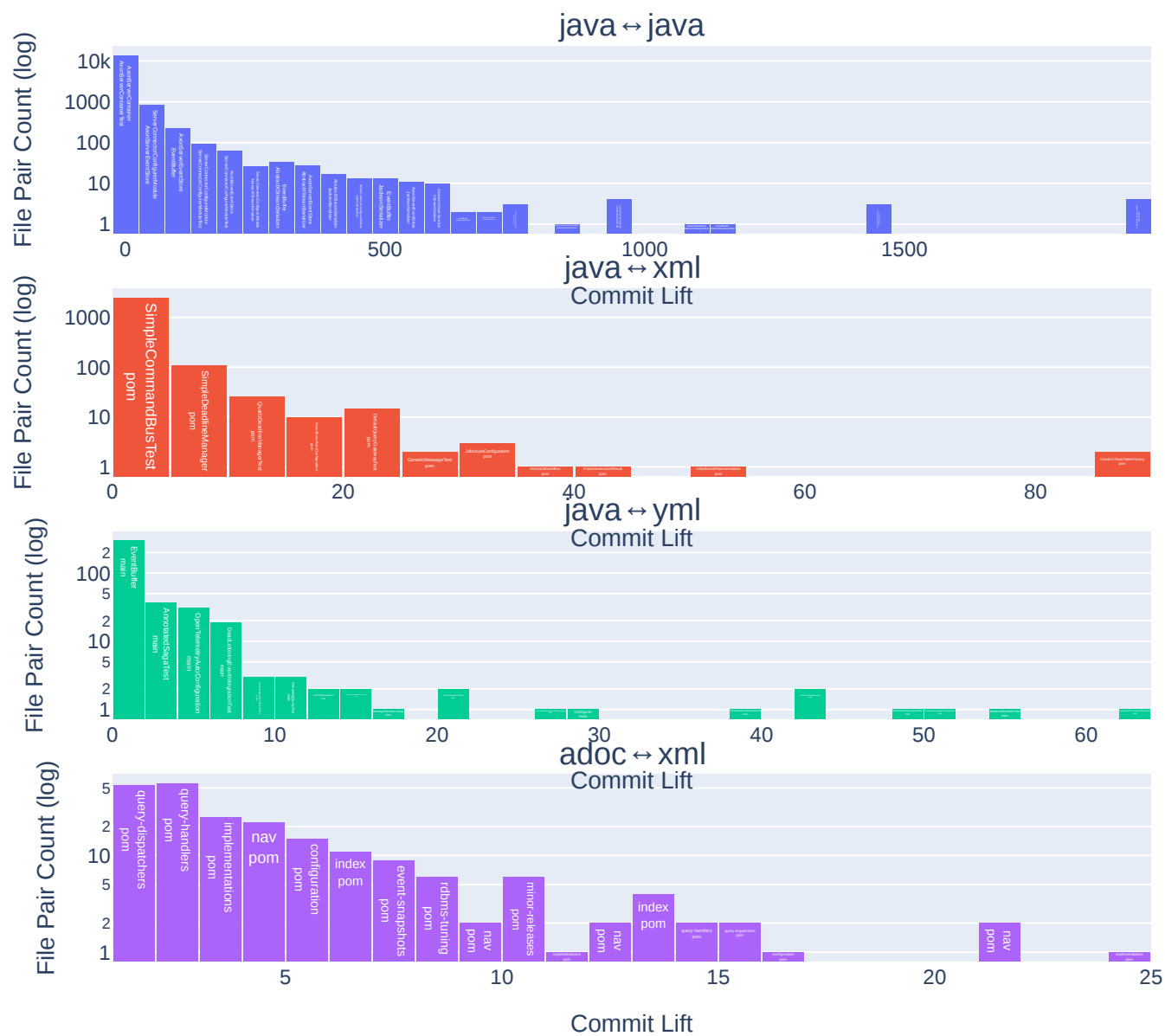
Co-Changed Files by their commit min confidence



Files changed together by commit lift

fileExtensionPair	updateCommitLift	GroupRank	
0	java↔java	1942.666667	1 AvroSchemaPackages↔Av
1	java↔java	1457.000000	2 DefaultEventProcessorSpanFactory↔DefaultEventProcessorS
2	java↔java	1165.600000	3 TimeoutWrappedMessageHandlingMemberTest↔UnitOfWorkTimeoul
3	java↔java	1092.750000	4 AxonServerEventStoreFactory↔AxonServerEventS
4	java↔java	971.333333	5 ClasspathAvroSchemaLoader↔SpecificRecordBaseClasspathAvrc
5	java↔java	832.571429	6 DeadlineDetailsSerializationTest↔JobRunnrDeadlineMan
6	java↔java	728.500000	7 EnqueuePolicy↔NoSuchDeadl
7	java↔java	699.360000	8 AxonServerNonTransientRemoteCommandHandlingException↔AxonServerNonTransientRemoteQueryHar
8	java↔java	693.809524	9 AvroSerializerStrategy↔ByteArrayToGenericR
9	java↔java	647.555556	10 GenericDeadLetter
10	java↔xml	86.726190	1 ContextAddingInterr
11	java↔xml	50.590278	2 UpstreamAwareStreamOb:
12	java↔xml	42.695971	3 JobRunnrEventSche
13	java↔xml	35.321212	4 EventStoreTestUt
14	java↔xml	33.996667	5 TestAggregateRoot↔benchmark-
15	java↔xml	33.167480	6 Message↔benchmark-
16	java↔xml	30.118863	7 JobRunnrEventSche
17	java↔xml	27.752381	8 MessagingCommandHandler↔benchmark-
18	java↔xml	26.076063	9 JobRunnrEventSche
19	java↔xml	24.283333	10 AbstractResourceInject
20	java↔yaml	62.357202	1 JobRunnrEventSchedi
21	java↔yaml	54.637500	2 AbstractResourceInjector↔
22	java↔yaml	50.879365	3 AvroSerializerAutoConfig
23	java↔yaml	48.566667	4 AxonTimeLimitedT
24	java↔yaml	43.753754	5 JobRunnrEventSchedulerTest↔slack-rele
25	java↔yaml	43.710000	6 ConfigurationScopeAwareProvider↔
26	java↔yaml	38.264646	7 AbstractDeadlineManagerTe
27	java↔yaml	29.028352	8 JobrunnrDeadlineMana
28	java↔yaml	26.848894	9 AbstractDeadlineManagerTestSuite↔slack-rele
29	java↔yaml	21.479115	10 JpaEventStoreAutoConfigurationWithoutAxonServerTest↔slack-rele
30	adoc↔xml	24.016484	1
31	adoc↔xml	21.347985	2
32	adoc↔xml	16.941860	3
33	adoc↔xml	15.059432	4 conf
34	adoc↔xml	14.667785	5
35	adoc↔xml	14.231990	6
36	adoc↔xml	13.723705	7 implem
37	adoc↔xml	13.185520	8 minor
38	adoc↔xml	13.038031	9 conf
39	adoc↔xml	12.008242	10 query

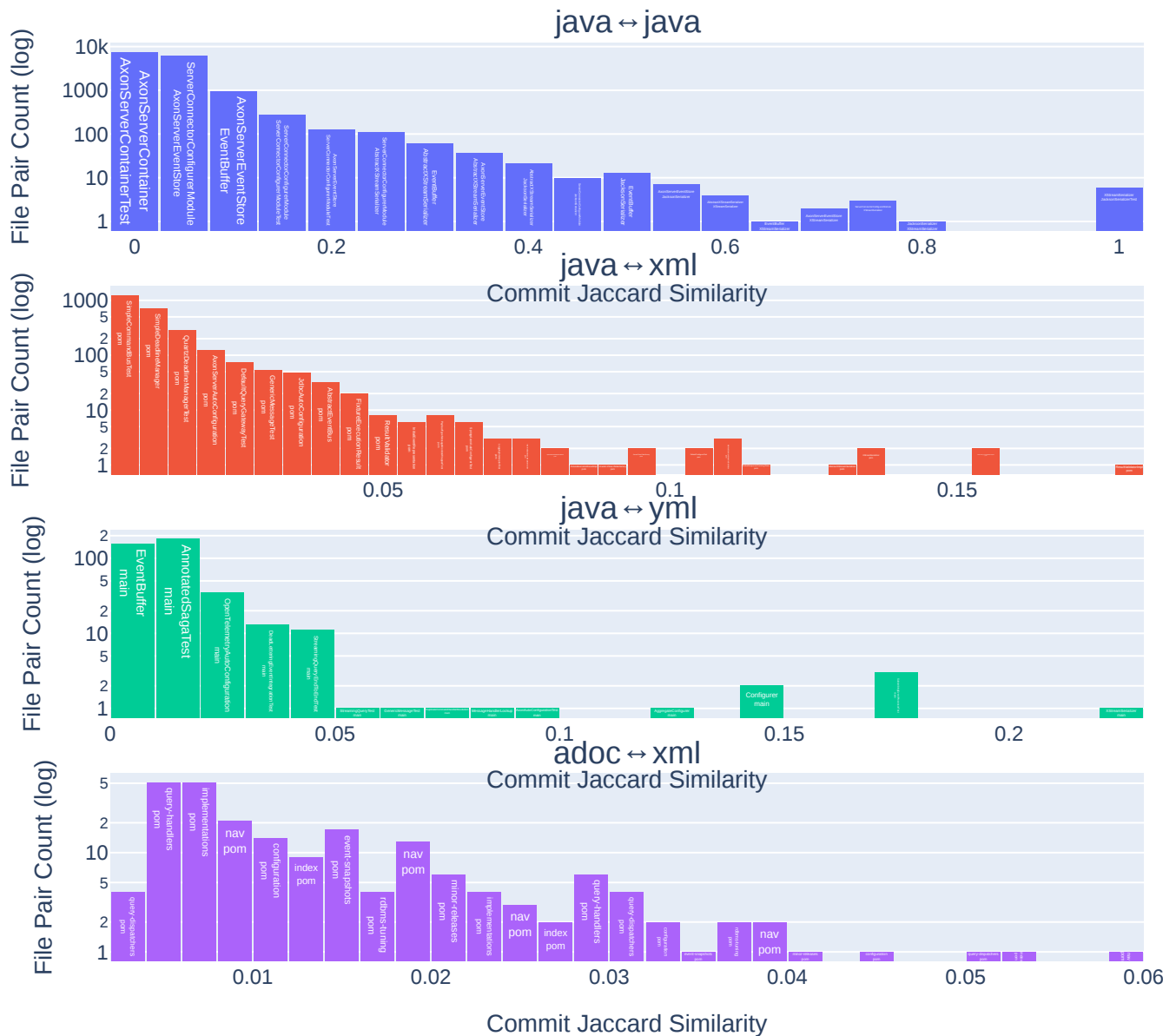
Co-Changed Files by their commit lift



Files changed together by commit Jaccard similarity

fileExtensionPair	updateCommitJaccardSimilarity	GroupRank	filePair
0	java↔java	1.000000	1AxonTimeLimitedTask↔UnitOfWorkTimeoutInterceptor
1	java↔java	0.800000	2DoNotEnqueueTest↔ShouldEnqueueTest
2	java↔java	0.750000	3DefaultEventProcessorSpanFactory↔DefaultEventProcessorSpanFactoryTest
3	java↔java	0.714286	4AvroUtil↔SpecificRecordBaseSerializerStrategy
4	java↔java	0.700000	5DoNotEnqueue↔ShouldEnqueue
5	java↔java	0.625000	6AvroSerializerStrategy↔ByteArrayToGenericRecordConverter
6	java↔java	0.600000	7AxonTimeLimitedTask↔TimeoutWrappedMessageHandlingMember
7	java↔java	0.555556	8MessageAuthorizationDispatchInterceptor↔MessageAuthorizationDispatchInterceptorTest
8	java↔java	0.545455	9ShouldEnqueue↔DoNotEnqueueTest
9	java↔java	0.529412	10AnnotatedChildEntity↔ChildEntity
10	java↔xml	0.180000	1JobRunrEventSchedulerTest↔pom
11	java↔xml	0.156250	2ContextAddingInterceptor↔logback
12	java↔xml	0.137405	3JobrunrDeadlineManagerTest↔pom
13	java↔xml	0.134454	4AbstractDeadlineManagerTestSuite↔pom
14	java↔xml	0.130435	5JobRunrEventSchedulerTest↔pom
15	java↔xml	0.113924	6JobRunrEventSchedulerTest↔pom
16	java↔xml	0.111111	7TestAggregateRoot↔benchmark-mongo-context
17	java↔xml	0.109375	8Message↔benchmark-mongo-context
18	java↔xml	0.108974	9AbstractDeadlineManagerTestSuite↔pom
19	java↔xml	0.106509	10JobrunrDeadlineManagerTest↔pom
20	java↔yaml	0.220339	1JobRunrEventSchedulerTest↔antora
21	java↔yaml	0.177419	2AvroSerializerAutoConfiguration↔antora
22	java↔yaml	0.174419	3JobRunrEventSchedulerTest↔slack-release-notification
23	java↔yaml	0.171053	4AbstractDeadlineManagerTestSuite↔antora
24	java↔yaml	0.145631	5AbstractDeadlineManagerTestSuite↔slack-release-notification
25	java↔yaml	0.144444	6JobrunrDeadlineManagerTest↔antora
26	java↔yaml	0.128205	7JobrunrDeadlineManagerTest↔slack-release-notification
27	java↔yaml	0.090909	8AbstractResourceInjector↔add-to-project
28	java↔yaml	0.081081	9ConfigurationScopeAwareProvider↔add-to-project
29	java↔yaml	0.071429	10AnnotatedMessageHandlingMemberTest↔antora
30	adoc↔xml	0.059322	1minor-releases↔pom
31	adoc↔xml	0.053571	2nav↔pom
32	adoc↔xml	0.051282	3rdbms-tuning↔pom
33	adoc↔xml	0.044872	4minor-releases↔pom
34	adoc↔xml	0.040000	5nav↔pom
35	adoc↔xml	0.039773	6minor-releases↔pom
36	adoc↔xml	0.038710	7rdbms-tuning↔pom
37	adoc↔xml	0.036232	8rdbms-tuning↔pom
38	adoc↔xml	0.036036	9event-snapshots↔pom
39	adoc↔xml	0.035294	10nav↔pom

Co-Changed Files by their commit jaccard similarity



Find pairwise changed files with many highly ranked metrics

Find those pairwise changed files that have a high rank in many metrics by calculating a combined (weighted) score based on the ranks of each metric. This is useful to identify pairs of files that score high in most metrics, which indicates a strong co-change relationship.

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCountExter
0	adoc↔xml	index↔pom	24	
1	adoc↔xml	nav↔pom	27	
2	adoc↔xml	nav↔pom	28	
3	adoc↔xml	configuration↔pom	28	
4	adoc↔xml	minor-releases↔pom	29	
5	adoc↔xml	rdbms-tuning↔pom	35	
6	adoc↔xml	index↔pom	38	
7	adoc↔xml	nav↔pom	39	
8	adoc↔xml	rdbms-tuning↔pom	41	
9	adoc↔xml	minor-releases↔pom	41	
10	java↔java	TimeoutWrappedMessageHandlingMemberTest↔UnitOfWorkTimeoutInterceptorTest	83	
11	java↔java	AvroSchemaPackages↔AvroSchemaScan	83	
12	java↔java	AvroSchemaScan↔ClasspathAvroSchemaLoader	83	
13	java↔java	AvroSchemaPackages↔ClasspathAvroSchemaLoader	83	
14	java↔java	MetadataContextGetter↔MetadataContextSetter	83	
15	java↔java	DefaultEventProcessorSpanFactory↔DefaultEventProcessorSpanFactoryTest	86	
16	java↔java	NoSuchDeadLetterException↔IgnoreTest	86	
17	java↔java	SpanUtils↔SpanUtilsTest	86	
18	java↔java	DoNotEnqueueTest↔ShouldEnqueueTest	92	
19	java↔java	AvroSerializerStrategy↔ByteArrayToGenericRecordConverter	99	
20	java↔xml	JobRunrEventSchedulerTest↔pom	36	
21	java↔xml	JobRunrEventSchedulerTest↔pom	44	
22	java↔xml	JobRunrEventSchedulerTest↔pom	47	
23	java↔xml	JobRunrEventSchedulerTest↔pom	62	
24	java↔xml	AbstractDeadlineManagerTestSuite↔pom	72	
25	java↔xml	ContextAddingInterceptor↔logback	74	
26	java↔xml	TokenAddingInterceptor↔logback	74	
27	java↔xml	JobRunrEventSchedulerTest↔pom	80	
28	java↔xml	JobRunrDeadlineManagerTest↔pom	84	
29	java↔xml	UpstreamAwareStreamObserver↔logback	84	
30	java↔yaml	JobRunrEventSchedulerTest↔antora	8	
31	java↔yaml	JobRunrEventSchedulerTest↔slack-release-notification	11	
32	java↔yaml	AvroSerializerAutoConfiguration↔antora	14	
33	java↔yaml	AbstractDeadlineManagerTestSuite↔slack-release-notification	22	
34	java↔yaml	AbstractDeadlineManagerTestSuite↔antora	23	
35	java↔yaml	JobRunrDeadlineManagerTest↔antora	27	
36	java↔yaml	AxonTimeLimitedTaskTest↔antora	33	
37	java↔yaml	JobRunrDeadlineManagerTest↔slack-release-notification	33	
38	java↔yaml	JobRunrEventSchedulerTest↔pullrequest	34	
39	java↔yaml	JobRunrEventSchedulerTest↔main	36	

Pairwise changed files with pareto-optimal metrics

A pair (count, confidence, jaccard, lift) is Pareto-optimal if there is no other pair that is better or equal in all metrics and strictly better in at least one. In other words, it is not "dominated" by any other pair.

The frontier = the “best tradeoffs.”

Pairwise changed files with pareto-optimal metrics - not considering file extensions

